

TECHNICAL SPECIFICATIONS

GRACE Hub

*Unified Nonprofit Operations Platform for the Detroit Public Schools Community District
Foundation*

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1. Purpose & Scope

This document defines the technical specifications for GRACE Hub, a custom unified operations platform for the Detroit Public Schools Community District Foundation ("the Foundation"). It is intended for the development team as the authoritative reference for architecture, functional requirements, data structures, integrations, and non-functional requirements.

Requirements are labeled with unique identifiers (e.g., PLT-01) for traceability through design, development, testing, and acceptance. Requirements use "must" for mandatory behavior and "should" for preferred behavior that may be negotiated during design review.

1.1 Product Vision

GRACE Hub serves as the Foundation's central operating system: it unifies departments on one platform, automates workflows, centralizes organizational knowledge, and provides leadership with real-time operational visibility.

1.2 In Scope

- Platform infrastructure: identity, permissions, workflow, notifications, search, AI framework, audit, storage
- Departmental modules: Executive/Board, Advancement (Donor CRM), Grants, Finance, Communications, Events, Programs, Operations, HR/Administration
- Integrations: Azure Entra ID, Microsoft 365, QuickBooks, Bill.com
- Organization-wide AI features and universal search

1.3 Out of Scope (Phase 4 / Future)

- White-label platform, multi-organization tenancy, subscription management, client administration, partner portals, public APIs, and a nonprofit module marketplace are documented as future architecture considerations (Section 12) but are not deliverables of the initial engagement.

2. System Architecture

2.1 Logical Architecture

The platform follows a modular, service-oriented architecture with the following layers:

- Presentation layer — responsive web application (mobile-first), rendering role-specific dashboards and module UIs
- Application layer — module services (CRM, Grants, Finance, Events, Programs, Operations, HR) built on shared platform services
- Shared platform services — authentication/authorization, workflow engine, notification engine, search/indexing, document management, AI service layer, audit service

- Data layer — relational system of record, document/blob storage, and search index
- Integration layer — connectors for Microsoft 365, QuickBooks, and Bill.com with queued, retryable sync jobs

2.2 Architectural Requirements

ARC-01 The system must be modular: each departmental module must be independently deployable or feature-flaggable without affecting other modules.

ARC-02 All modules must consume shared platform services (auth, workflow, notifications, search, documents, audit) rather than implementing their own.

ARC-03 The system must be cloud-hosted with environment separation for development, staging, and production.

ARC-04 All application functionality must be exposed through an internal API layer to support future public APIs and integrations (Phase 4).

ARC-05 The data model must anticipate multi-tenancy (organization ID scoping) even though the initial deployment is single-tenant.

ARC-06 The UI must be fully responsive and functional on mobile devices for all end-user workflows; administrative configuration may be desktop-optimized.

3. Identity, Access & Security

3.1 Authentication

SEC-01 The system must authenticate users exclusively via Azure Entra ID single sign-on (OIDC/SAML). No local passwords.

SEC-02 User provisioning and deactivation should sync from Entra ID (SCIM or scheduled directory sync); deactivated users must lose access within 15 minutes.

3.2 Authorization

SEC-03 The system must implement role-based access control with multi-level permissions: platform roles (admin, staff, board, read-only), department scopes, and record-level restrictions (e.g., restricted donor records, HR documents).

SEC-04 Permission checks must be enforced server-side on every API call; UI hiding alone is not sufficient.

SEC-05 Administrators must be able to manage users, roles, and department scopes through an admin UI without developer involvement.

3.3 Audit & Data Protection

SEC-06 The system must write an immutable audit log for all create, update, delete, permission-change, export, and login events, capturing actor, timestamp, record, and before/after values where practical.

SEC-07 Audit logs must be searchable by administrators and retained for a minimum of seven years.

SEC-08 All data must be encrypted in transit (TLS 1.2+) and at rest.

SEC-09 The system must support field-level sensitivity flags (e.g., donor wealth data, HR records) that restrict visibility to authorized roles.

SEC-10 Backups must run at least daily with a documented disaster recovery plan; targets: RPO ≤ 24 hours, RTO ≤ 8 hours (to be confirmed in design review).

4. Shared Platform Services

4.1 Workflow Engine

WFL-01 The system must provide a configurable workflow engine supporting sequential and parallel approval routing, conditional branching (e.g., by amount, department, record type), escalations on timeout, and status tracking.

WFL-02 Workflows must be definable by administrators through configuration, not code, for common patterns (approvals, review chains, task handoffs).

WFL-03 Every workflow instance must show its current step, assignee, history, and elapsed time, and must appear in the assignee's Task Center and the executive Pending Approvals view.

4.2 Notification Engine

NTF-01 The system must deliver notifications in-app and by email, with per-user preferences and digest options.

NTF-02 Notifications must be triggered by workflow events, deadlines (grants, reports, pledges), assignments, and mentions.

NTF-03 A Notification Center must aggregate all notifications with read/unread state and deep links to source records.

4.3 Document Management

DOC-01 The system must provide a central document repository with folders, tags, versioning, and permissions inherited from module records.

DOC-02 Documents must be attachable to any record type (donor, grant, event, vendor, board meeting, etc.) and discoverable through global search.

DOC-03 The repository must integrate with Microsoft 365 for viewing/editing Office documents where feasible.

4.4 Global Search

SRC-01 Universal search must cover donors, grants, events, communications, programs, vendors, board documents, templates, policies, financial reports, meeting notes, and contacts.

SRC-02 Search results must respect record-level permissions and return results in under 2 seconds for typical queries.

SRC-03 Search must support filters by record type, department, date range, and owner.

4.5 AI Service Layer

AIS-01 AI features must be delivered through a central AI service layer so models, prompts, and providers can be changed without modifying modules.

AIS-02 Foundation data must not be used to train third-party models; the AI provider's data-handling terms must be documented and approved.

AIS-03 All AI-generated content (proposals, reports, emails) must be presented as drafts requiring human review before send/submit; AI actions must be attributable in the audit log.

AIS-04 The AI knowledge assistant must ground responses in the Foundation's knowledge library and cite source documents.

4.6 Activity Feed & Calendar

PLT-01 A cross-module activity feed must show recent record activity scoped to the user's permissions.

PLT-02 A shared organizational calendar must aggregate events, grant deadlines, board meetings, and program dates, with department filters.

5. Core Data Model

The following entities form the shared system of record. Detailed field-level schemas will be produced during design; the relationships below are mandatory.

Entity	Description & Key Relationships
Contact	Person or organization (individual, corporate, foundation, org). Supports household grouping, relationship mapping between contacts, notes, attachments, and full communication history. Shared by Advancement, Communications, Events, and Programs.
Gift / Pledge	Financial contribution linked to a Contact and optionally a Campaign, Event, or Grant. Types: one-time, recurring, pledge (with schedule), sponsorship, tribute, matching, grant revenue.
Campaign	Fundraising campaign aggregating gifts, communications, and attribution.

Entity	Description & Key Relationships
Prospect Plan	Moves-management record on a Contact: stage (qualification, cultivation, solicitation, stewardship, renewal), proposal tracking, tasks, and assigned officer.
Funder / Grant	Funder profile plus grant records across the lifecycle: lead, opportunity, proposal, award, active, closed. Awards carry amount, dates, restrictions, payment schedule, agreements, milestones, reporting deadlines, and spenddown against budget.
Event	Event with venue, dates, budget, staff assignments, registrations/tickets, check-in, sponsors (packages, payments), vendors (contracts, payments), and financials.
Program	Program/cohort/initiative with participants (registration, attendance, outcomes), volunteers, and impact metrics (students, families, scholars, community members served).
Financial Record	Purchase orders, invoices, payments, and budget lines synchronized with QuickBooks and Bill.com; linked to vendors, grants, events, and departments for reporting.
Task / Project	Assignable task with due date, project grouping, and workflow linkage; surfaces in Task Center.
Board Record	Board meetings, agendas, packets, minutes, governance documents, members, and committees.
Knowledge Asset	Policies, SOPs, templates, boilerplate text, forms, and training content with tags and permissions.
Communication	Logged email, call, meeting, or campaign touch tied to Contacts and Campaigns, with engagement metrics.

Cross-cutting requirements:

DAT-01 Every entity must carry created/modified metadata, owner, department, and permission scope.

DAT-02 Impact metrics (students, families, scholars, community members served; outcomes) must be recordable against Grants, Programs, and Events and roll up to organization-level dashboards.

DAT-03 The system must support data import from existing spreadsheets and legacy systems, with deduplication on Contacts (name/email/address matching) during migration.

6. Module Specifications

6.1 Executive Leadership & Board

EXE-01 Executive dashboard must display organizational KPIs, department scorecards, financial snapshot, grant snapshot, fundraising dashboard, event dashboard, community impact dashboard, pending approvals, and the organizational calendar, with drill-down to source records.

EXE-02 Reporting suite must generate board, executive, KPI, department, impact, budget, grant, and fundraising reports, exportable to PDF and Excel, with scheduled delivery.

EXE-03 Board packet builder must assemble agendas, reports, and documents into a paginated packet (PDF) with a wizard-driven flow; packets, minutes, and governance documents must be stored in a permissioned board library.

EXE-04 Board member directory and committee management must track membership, terms, and committee assignments.

6.2 Advancement — Donor CRM

ADV-01 Contact management must support all contact types with household management, relationship mapping, contact history, notes, and attachments (see Data Model).

ADV-02 Giving must record all gift types with receipts/acknowledgment tracking; recurring gifts and pledge schedules must generate expected-payment records and reminders.

ADV-03 Prospect management must implement moves-management stages with proposal tracking, stewardship actions, and renewal reminders.

ADV-04 Communications log must capture email history, call history, meeting notes, follow-up tasks, and reminders on the contact timeline.

ADV-05 Analytics must compute donor retention, LYBUNT, SYBUNT, average gift, lifetime giving, and pipeline value, filterable by year, campaign, and segment.

6.3 Grants

GRT-01 Pipeline must track leads and opportunities against a funder database, with deadline tracking and automated reminders.

GRT-02 Proposal development must support status, collaborators, tasks, budget attachments, and reuse from a narrative/boilerplate library.

GRT-03 Awards must capture amount, dates, restrictions, payment schedule, and executed agreements.

GRT-04 Active grant management must track milestones, reporting deadlines, communications, and spenddown against budget with variance alerts.

GRT-05 AI assistance must draft proposals, reports, and letters grounded in the knowledge library and prior applications (subject to AIS requirements).

GRT-06 Impact tracking must record students, families, scholars, and community members served, plus outcomes and success metrics per grant.

6.4 Finance

FIN-01 Financial dashboard must display cash on hand, revenue, expenses, AP, AR, net assets, and budget vs. actual, sourced from QuickBooks sync.

FIN-02 Purchasing must support purchase orders, invoice upload (with OCR-assisted data capture where feasible), configurable approval routing, payment scheduling, three-way match (PO / receipt / invoice), vendor payments via Bill.com, and exception handling queues.

FIN-03 Financial reports must include P&L, balance sheet, budget reports, department expenses, grant expenses, and revenue by source.

FIN-04 Expenses must be attributable to departments, grants, and events to support restricted-fund and grant spenddown reporting.

6.5 Communications

COM-01 Email campaigns must support templates, audience segmentation, and distribution lists built from CRM queries.

COM-02 Engagement tracking must capture open and click rates and write them back to contact timelines and campaign attribution.

COM-03 Event invitations must link campaigns to Events and track responses.

COM-04 AI assistance must draft emails and newsletters and suggest campaigns (subject to AIS requirements).

6.6 Events

EVT-01 Planning must manage events, venues, dates, budgets, and staff assignments.

EVT-02 Registration must support ticket sales, registration forms, check-in (including mobile check-in), and attendance capture.

EVT-03 Sponsor management must track packages, commitments, and payments; vendor management must track contracts and payments.

EVT-04 Event reporting must show revenue, expenses, attendance, ROI, and year-over-year historical comparison.

6.7 Programs

PRG-01 Program management must support programs, cohorts, initiatives, community events, and volunteers.

PRG-02 Participant tracking must cover registration, attendance, outcomes, and impact metrics.

PRG-03 Reporting must present program KPIs, attendance, and community impact, rolling up to executive dashboards.

6.8 Operations

OPS-01 Task management must support tasks, projects, assignments, due dates, and notifications, aggregated in a Task Center.

OPS-02 Workflow automation must provide approval routing, status tracking, escalations, and smart notifications (via the shared workflow engine).

OPS-03 Knowledge Center must house policies, SOPs, templates, HR documents, forms, and a resource library with tagging, versioning, and permissions.

6.9 HR / Administration

HRA-01 Employee directory must sync from Entra ID and track roles, permissions, and training records.

HRA-02 Learning Center must host training videos, help articles, FAQs, and platform documentation, searchable and role-filtered.

7. Integration Specifications

System	Direction & Data	Notes
Azure Entra ID	Inbound: authentication (SSO), user provisioning, group-to-role mapping	OIDC preferred; deactivation propagates ≤ 15 min
Microsoft 365	Bi-directional: calendar events, email send/log, document viewing/editing (SharePoint/OneDrive), meeting artifacts	Graph API; respect M365 permissions
QuickBooks	Bi-directional: chart of accounts, transactions, budgets in; invoices/bills and class/departments coding out	Scheduled sync $\leq$ hourly; reconciliation report for sync errors
Bill.com	Bi-directional: vendors, bills, approval status, payment status	Approval state must reconcile with GRACE Hub workflow to avoid dual approval conflicts

Integration requirements:

INT-01 All integrations must be queued and retryable, with an admin-visible sync log, error queue, and alerting on repeated failures.

INT-02 A single source of truth must be designated per data domain (e.g., QuickBooks for the general ledger, GRACE Hub for donors/grants) and documented in the design phase.

INT-03 Integration credentials must be stored in a secrets manager, never in code or configuration files.

8. Non-Functional Requirements

Category	Requirement
Performance	Page loads ≤ 3 s on standard broadband; dashboards ≤ 5 s; search ≤ 2 s; imports and reports may run asynchronously with progress indication.
Availability	99.5% uptime target excluding scheduled maintenance; maintenance windows communicated in advance.
Scalability	Support 100+ concurrent users and 10 years of transactional history without degradation; architecture must scale to multi-organization use (Phase 4).
Accessibility	WCAG 2.1 AA for all end-user interfaces.
Browser support	Current versions of Edge, Chrome, Safari, and Firefox; iOS and Android mobile browsers.
Data retention	Audit logs ≥ 7 years; soft-delete with 90-day recovery for user-deleted records.
Localization	US English at launch; currency in USD; date/time in ET with user-level preference.

9. Reporting & Analytics Framework

RPT-01 A shared reporting service must power all module reports with consistent filters (date range, department, campaign, grant, program) and export to PDF and Excel.

RPT-02 Scheduled reports must be deliverable by email to defined recipients (e.g., monthly board financial summary).

RPT-03 Dashboards must refresh from live data or near-real-time caches (≤ 15 minutes stale).

RPT-04 Phase 3 must add predictive dashboards, automated report generation, workflow intelligence, executive insights, advanced analytics, and organization benchmarking on top of the same reporting service.

10. Delivery Phases

Phase	Scope	Duration
Phase 1 — Core Infrastructure	Entra SSO & M365 integration; user/role management; central dashboard; global search; document repository; grants (core); donor CRM (core); QuickBooks & Bill.com integrations; communications (core); workflow engine; basic reporting; audit logs	12–16 weeks
Phase 2 — Operational Modules	Events; board packet wizard; programs; advanced finance workflows; advanced grant management;	8–12 weeks

Phase	Scope	Duration
	donor moves management; cross-department collaboration; AI-assisted content creation; mobile optimization; Knowledge Center	
Phase 3 — Intelligence & Automation	AI assistant; AI search; predictive dashboards; automated reporting; workflow intelligence; executive insights; advanced analytics; benchmarking	8–10 weeks
Phase 4 — Enterprise Expansion	White-label, multi-org tenancy, subscriptions, client admin, partner portals, additional ERP/CRM integrations, public APIs, module marketplace	Future

10.1 MVP Definition

The first production release must include: platform infrastructure and security; central dashboard and executive reporting; donor CRM and grant management; finance integrations and approval workflows; document management and global search; and communications with workflow automation.

11. Testing & Acceptance

QAT-01 Each requirement ID in this document must map to at least one test case; a traceability matrix must be maintained.

QAT-02 Testing must include unit, integration (including all third-party sync scenarios and failure modes), permission/security testing, and user acceptance testing with Foundation staff per module.

QAT-03 Acceptance per phase requires: all "must" requirements passing, no open critical or high defects, completed security review, and sign-off by the Foundation project owner.

QAT-04 A staging environment seeded with representative (anonymized) data must be available for UAT.

12. Future Architecture Considerations (Phase 4)

The initial build must not preclude the following planned expansions. Design decisions that would block them require explicit sign-off:

- Multi-organization tenancy with white-label theming and per-tenant configuration
- Subscription management and client administration for external nonprofit customers
- Partner portals with externally scoped authentication
- Public APIs with versioning, rate limiting, and API-key management
- Additional ERP/CRM integrations and a marketplace for nonprofit modules

Appendix A — Open Decisions

- Technology stack selection (framework, database, hosting platform) — to be proposed by the development team and approved at design review
- AI model provider(s) and data-processing agreement
- Final RPO/RTO targets and hosting region
- Data migration source inventory and cutover plan
- Email sending infrastructure for campaigns (native vs. integrated ESP) and deliverability ownership